Experiment No. 1

1. Traversal (Display Array)

```
#include<stdio.h>

int main()
{
   int arr[100], n, i;

   printf("Enter number of elements: ");
   scanf("%d",&n);


   printf("Enter elements:\n");
   for(i=0;i<n;i++)
      scanf("%d",&arr[i]);

   printf("Array elements are:\n");
   for(i=0;i<n;i++)
      printf("%d ",arr[i]);

   return 0;
}
```

```
cticals\1st practical\"Traversal
Enter number of elements: 5
Enter elements:
23
54
11
23
5
Array elements are:
23 54 11 23 5
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>
```

2. Insertion

```
#include<stdio.h>

int main()
{
   int arr[100], n, i, pos, val;

   printf("Enter number of elements: ");
   scanf("%d",&n);

   for(i=0;i<n;i++)
      scanf("%d",&arr[i]);
```

```
    printf("Enter position: ");
    scanf("%d",&pos);

    printf("Enter value: ");
    scanf("%d",&val);

    for(i=n;i>=pos;i--)
        arr[i]=arr[i-1];

    arr[pos-1]=val;
    n++;

    printf("Array after
insertion:\n");
    for(i=0;i<n;i++)
        printf("%d ",arr[i]);

    return 0;
}
```

```
cticals\1st practical\"Insertion
Enter number of elements: 5
34
23
12
43
22
Enter position: 3
Enter value: 24
Array after insertion:
34 23 24 12 43 22
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>
```

3. Deletion

```
#include<stdio.h>

int main()
{
    int arr[100], n, i, pos;

    printf("Enter number of elements:
");
    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&arr[i]);

    printf("Enter position to delete: ");
    scanf("%d",&pos);

    for(i=pos-1;i<n-1;i++)
        arr[i]=arr[i+1];

    n--;

    printf("Array after deletion:\n");
    for(i=0;i<n;i++)
```

```
icals\1st practical\"deletion
Enter number of elements: 10
12
43
45
76
23
454
56
34
67
3
Enter position to delete: 4
Array after deletion:
12 43 45 23 454 56 34 67 3
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>
```

```
    printf("%d ",arr[i]);

  return 0;
}
```

4. Searching (Linear Search)

```
#include<stdio.h>

int main()
{
  int arr[100], n, i, key, found=0;

  printf("Enter number of elements: ");
  scanf("%d",&n);

  for(i=0;i<n;i++)
    scanf("%d",&arr[i]);

  printf("Enter element to search: ");
  scanf("%d",&key);

  for(i=0;i<n;i++)
  {
    if(arr[i]==key)
    {
      printf("Element found at position %d",i+1);
      found=1;
      break;
    }
  }

  if(found==0)
    printf("Element not found");

  return 0;
}
```

```
cticals\1st practical\"searching
Enter number of elements: 10
45
43
23
12
87
98
34
65
2
45
Enter element to search: 87
Element found at position 5
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>
```

5. Sorting (Ascending Order)

```
#include<stdio.h>

int main()
{
  int arr[100], n, i, j, temp;
```

```
    printf("Enter number of elements: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&arr[i]);

    for(i=0;i<n-1;i++)
    {
        for(j=i+1;j<n;j++)
        {
            if(arr[i]>arr[j])
            {
                temp=arr[i];
                arr[i]=arr[j];
                arr[j]=temp;
            }
        }
    }

    printf("Sorted array:\n");
    for(i=0;i<n;i++)
        printf("%d ",arr[i]);

    return 0;
}
```

```
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>cd "c:\Users
als\1st practical\"sorting
Enter number of elements: 10
12
23
34
45
32
46
67
89
23
56
Sorted array:
12 23 23 32 34 45 46 56 67 89
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>
```

6. Merging Two Arrays

```
#include<stdio.h>

int main()
{
    int a[50], b[50], c[100], n1, n2, i;

    printf("Enter size of first array: ");
    scanf("%d",&n1);

    for(i=0;i<n1;i++)
        scanf("%d",&a[i]);

    printf("Enter size of second array: ");
    scanf("%d",&n2);

    for(i=0;i<n2;i++)
        scanf("%d",&b[i]);
```

```
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>cd "c:\User
icals\1st practical\"mearging
Enter size of first array: 5
34
23
43
654
76
Enter size of second array: 5
12
23
34
54
65
Merged array:
34 23 43 654 76 12 23 34 54 65
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>
```

```
    for(i=0;i<n1;i++)
        c[i]=a[i];

    for(i=0;i<n2;i++)
        c[n1+i]=b[i];

    printf("Merged array:\n");
    for(i=0;i<n1+n2;i++)
        printf("%d ",c[i]);

    return 0;
}
```

7. Reversing Array

```
#include<stdio.h>

int main()
{
    int arr[100], n, i;

    printf("Enter number of elements: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&arr[i]);

    printf("Reversed array:\n");
    for(i=n-1;i>=0;i--)
        printf("%d ",arr[i]);

    return 0;
}
```

```
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>cd "c:\Us
cticals\1st practical\"reversing
Enter number of elements: 10
23
34
54
1
34
45
56
76
98
43
Reversed array:
43 98 76 56 45 34 1 54 34 23
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>
```

8. Copying Array

```
#include<stdio.h>
int main()
    int a[100], b[100], n, i;
    printf("Enter number of elements:
    scanf("%d",&n);
    for(i=0;i<n;i++)
scanf("%d",&a[i]);
    for(i=0;i<n;i++)
b[i]=a[i];
    printf("Copied array:\n");
```

```
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>cd "c:\Users
t practical\"copy
Enter number of elements: 10
45
34
32
78
11
23
90
87
65
54
Copied array:
45 34 32 78 11 23 90 87 65 54
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>
```

```
for(i=0;i<n;i++)
      printf("%d ",b[i]);

   return 0;
}
```

9. Sum and Average

```
#include<stdio.h>

int main()
{
   int arr[100], n, i, sum=0;
   float avg;

   printf("Enter number of
elements: ");
   scanf("%d",&n);

   for(i=0;i<n;i++)
      scanf("%d",&arr[i]);

   for(i=0;i<n;i++)
      sum=sum+arr[i];

   avg = sum/n;

   printf("Sum = %d\n",sum);
   printf("Average = %.2f",avg);

   return 0;
}
```

```
als\1st practical\"sum_avg
Enter number of elements: 5
12
23
34
45
56
Sum = 170
Average = 34.00
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>
```

10. Maximum and Minimum

```
#include<stdio.h>

int main()
{
   int arr[100], n, i, max, min;

   printf("Enter number of elements: ");
   scanf("%d",&n);

   for(i=0;i<n;i++)
      scanf("%d",&arr[i]);
```

```
    max=min=arr[0];

    for(i=1;i<n;i++)
    {
        if(arr[i]>max)
            max=arr[i];

        if(arr[i]<min)
            min=arr[i];
    }

    printf("Maximum = %d\n",max);
    printf("Minimum = %d",min);

    return 0;
}
```

```
als\1st practical\"min_max
Enter number of elements: 10
34
45
65
11
87
2
09
88
98
56
Maximum = 98
Minimum = 2
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>
```

11. Second Largest Element

```
#include<stdio.h>

int main()
{
    int arr[100], n, i, largest, second;

    printf("Enter number of elements:
");
    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&arr[i]);

    largest = second = arr[0];

    for(i=1;i<n;i++)
    {
        if(arr[i]>largest)
        {
            second = largest;
            largest = arr[i];
        }
        else if(arr[i]>second && arr[i]!=largest)
        {
            second = arr[i];
        }
    }
```

```
 practicals\1st practical\"2nd_largest
Enter number of elements: 10
34
32
56
76
545
23
4
23
98
67
Second largest = 98
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>
```

```
    printf("Second largest = %d",second);

    return 0;
}
```

12. Second Minimum Element

```
#include<stdio.h>

int main()
{
    int arr[100], n, i, min, second;

    printf("Enter number of elements: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&arr[i]);

    min = second = arr[0];


    for(i=1;i<n;i++)
    {
        if(arr[i]<min)
        {
            second = min;
            min = arr[i];
        }
        else if(arr[i]<second && arr[i]!=min)
        {
            second = arr[i];
        }
    }

    printf("Second minimum = %d",second);

    return 0;
}
```

```
als\1st practical\"2nd_min
Enter number of elements: 10
343
232
54
56
676
1
23
45
22
66
Second minimum = 22
c:\Users\PRASANNA\Desktop\DS practicals\1st practical>
```